Introduction to Cryptography

Lecture: Prime Numbers in Cryptography

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Why Study Prime Numbers?

Prime numbers are one of the central topics of number theory.

They are especially important in cryptography because many cryptographic techniques depend on arithmetic involving large prime numbers.

Key Idea

Prime numbers act like the basic building blocks of positive integers.

What is a Divisor?

An integer a is said to be a divisor of an integer b if:

$$b = ak$$

for some integer k.
We write:

$$a \mid b$$

and say:

"a divides b"

Examples of Divisibility

Consider:

$$12 \mid 36$$

because:

$$36 = 12 \times 3$$

Similarly:

$$5 \mid 40$$

because:

$$40 = 5 \times 8$$

But:

$$7 \nmid 20$$

What is a Prime Number?

Definition

An integer $p > 1$ is called a **prime number** if its only positive divisors are:

$$1 \quad \text{and} \quad p$$

Examples:

$$2,\ 3,\ 5,\ 7,\ 11,\ 13,\ 17,\ 19, \ldots$$

Prime vs Composite Numbers

Prime

Has exactly two positive divisors.
Examples:

$$2,\ 3,\ 5,\ 7,\ 11$$

Composite

Has more than two positive divisors.
Examples:

$$4,\ 6,\ 8,\ 9,\ 10$$

Is 1 a Prime Number?

No.

1 is neither prime nor composite

A prime number must have exactly two positive divisors:

1 and itself

But the number 1 has only one positive divisor.

Some Small Prime Numbers

The prime numbers below 100 are:

2, 3, 5, 7, 11, 13, 17, 19, 23, 29,
31, 37, 41, 43, 47, 53, 59, 61, 67, 71,
73, 79, 83, 89, 97

Observation

2 is the only even prime number.

Example: Is 13 Prime?

Possible positive divisors of 13 are:

$$1,\ 13$$

There are no other integers that divide 13 exactly.
Therefore:

$$\boxed{13 \text{ is prime}}$$

Example: Is 15 Prime?

The positive divisors of 15 are:

$$1,\ 3,\ 5,\ 15$$

Since 15 has divisors other than 1 and itself:

$$\boxed{\text{15 is composite}}$$

Also:

$$15 = 3 \times 5$$

Fundamental Theorem of Arithmetic

Theorem

Every positive integer greater than 1 can be expressed uniquely as a product of prime numbers, apart from the ordering of the factors.

General form:

$$\boxed{a = p_1^{a_1} p_2^{a_2} \cdots p_t^{a_t}}$$

where:

$$p_1 < p_2 < \cdots < p_t$$

are prime numbers and each exponent is a positive integer.

Prime Factorization Example: 91

Consider:

$$91$$

We can factor it as:

$$91 = 7 \times 13$$

Both 7 and 13 are prime.
Therefore:

$$\boxed{91 = 7^1 \times 13^1}$$

Prime Factorization Example: 3600

Start with:

$$3600$$

Its prime factorization is:

$$3600 = 2^4 \times 3^2 \times 5^2$$

Verification:

$$2^4 = 16$$

$$3^2 = 9$$

$$5^2 = 25$$

Thus:

Prime Factorization Example: 11011

Consider:

$$11011$$

Its prime factorization is:

$$\boxed{11011 = 7 \times 11^2 \times 13}$$

Verification:

$$11^2 = 121$$

$$7 \times 121 \times 13 = 11011$$

Representation Using All Prime Numbers

Let P represent the set of all prime numbers.
Then any positive integer a can be written as:

$$\boxed{a = \prod_{p \in P} p^{a_p}}$$

where:

$$a_p \geq 0$$

For any particular integer, most of the exponents a_p are zero.

Representing 12 Using Prime Exponents

Prime factorization:

$$12 = 2^2 \times 3^1$$

Therefore:

$$a_2 = 2$$

$$a_3 = 1$$

All other prime exponents are zero.
So 12 may be represented by:

$$\boxed{\{a_2 = 2,\ a_3 = 1\}}$$

Representing 18 Using Prime Exponents

Prime factorization:

$$18 = 2^1 \times 3^2$$

Therefore:

$$a_2 = 1$$

$$a_3 = 2$$

Hence:

$$\boxed{\{a_2 = 1,\ a_3 = 2\}}$$

Representing 91 Using Prime Exponents

We know:

$$91 = 7 \times 13$$

Therefore:

$$a_7 = 1$$

$$a_{13} = 1$$

Thus:

$$\boxed{\{a_7 = 1,\ a_{13} = 1\}}$$

Multiplication Using Prime Exponents

Suppose:

$$a = \prod_{p \in P} p^{a_p}$$

and

$$b = \prod_{p \in P} p^{b_p}$$

If:

$$k = ab$$

then:

$$\boxed{k_p = a_p + b_p}$$

for every prime p.

Example: Multiply 12 and 18

Prime factorizations:

$$12 = 2^2 \times 3$$

$$18 = 2 \times 3^2$$

Now:

$$12 \times 18 = (2^2 \times 3)(2 \times 3^2)$$

Add corresponding exponents:

$$2^{2+1} \times 3^{1+2}$$

$$= 2^3 \times 3^3$$

Multiplication Example: Final Result

We obtained:

$$2^3 \times 3^3$$

Therefore:

$$8 \times 27$$

$$= 216$$

Hence:

$$\boxed{12 \times 18 = 216}$$

and:

$$216 = 2^3 \times 3^3$$

Understanding the Exponents

For:

$$12 = 2^2 \times 3^1$$

and:

$$18 = 2^1 \times 3^2$$

the product has:

$$k_2 = 2 + 1 = 3$$

and:

$$k_3 = 1 + 2 = 3$$

Therefore:

$$216 = 2^3 \times 3^3$$

Divisibility Using Prime Factors

Suppose:

$$a = \prod_{p \in P} p^{a_p}$$

and:

$$b = \prod_{p \in P} p^{b_p}$$

Then:

$$\boxed{a \mid b}$$

if:

$$\boxed{a_p \leq b_p \quad \text{for every prime } p}$$

Why Does the Divisibility Rule Work?

Consider a prime power:

$$p^n$$

It can be divided exactly by:

$$p^0, p^1, p^2, \ldots, p^n$$

but not by:

$$p^{n+1}$$

Therefore, every prime exponent in the divisor must be less than or equal to the corresponding exponent in the number being divided.

Example: Does 12 Divide 36?

Prime factorizations:

$$12 = 2^2 \times 3$$

$$36 = 2^2 \times 3^2$$

Compare the exponents.
For prime 2:

$$2 \leq 2$$

For prime 3:

$$1 \leq 2$$

Therefore:

$$\boxed{12 \mid 36}$$

Example: Does 18 Divide 24?

Prime factorizations:

$$18 = 2^1 \times 3^2$$

$$24 = 2^3 \times 3^1$$

For prime 2:

$$1 \leq 3$$

but for prime 3:

$$2 \not\leq 1$$

Therefore:

$$\boxed{18 \nmid 24}$$

Greatest Common Divisor

The greatest common divisor of a and b is the largest positive integer that divides both. It is written as:

$$\gcd(a, b)$$

If:

$$k = \gcd(a, b)$$

then using prime exponents:

$$\boxed{k_p = \min(a_p, b_p)}$$

for every prime p.

Example: $\gcd(18, 300)$

Prime factorization of 18:

$$18 = 2^1 \times 3^2$$

Prime factorization of 300:

$$300 = 2^2 \times 3^1 \times 5^2$$

Take the minimum exponent for each prime.

Calculating $\gcd(18, 300)$

For prime 2:

$$\min(1, 2) = 1$$

For prime 3:

$$\min(2, 1) = 1$$

For prime 5:

$$\min(0, 2) = 0$$

Therefore:

$$\gcd(18, 300) = 2^1 \times 3^1 \times 5^0$$

$$\boxed{\gcd(18, 300) = 6}$$

Three Important Prime-Exponent Rules

Multiplication

$$(ab)_p = a_p + b_p$$

Divisibility

$$a \mid b \quad \Longrightarrow \quad a_p \leq b_p$$

Greatest Common Divisor

$$\gcd(a, b)_p = \min(a_p, b_p)$$

Numerical Practice 1

Find the prime factorization of:

$$72$$

Numerical Practice 1

Find the prime factorization of:

$$72$$

Start dividing:

$$72 = 2 \times 36$$

$$= 2 \times 2 \times 18$$

$$= 2 \times 2 \times 2 \times 9$$

$$= 2^3 \times 3^2$$

Therefore:

$$\boxed{72 = 2^3 \times 3^2}$$

Numerical Practice 2

Find the prime factorization of:

$$180$$

Numerical Practice 2

Find the prime factorization of:

$$180$$

$$180 = 18 \times 10$$

$$= (2 \times 3^2)(2 \times 5)$$

Therefore:

$$\boxed{180 = 2^2 \times 3^2 \times 5}$$

Numerical Practice 3: GCD

Find:

$$\gcd(72, 120)$$

Prime factorization:

$$72 = 2^3 \times 3^2$$

$$120 = 2^3 \times 3 \times 5$$

Minimum exponents:

$$2^3 \times 3^1$$

Therefore:

$$\boxed{\gcd(72, 120) = 24}$$

Classroom Activity

Determine whether each number is prime or composite.

1. 17
2. 21
3. 29
4. 35
5. 43
6. 51

Also give the prime factorization of every composite number.

Classroom Activity: Answers

17 Prime

$21 = 3 \times 7$

29 Prime

$35 = 5 \times 7$

43 Prime

$51 = 3 \times 17$

An Important Computational Observation

Prime factorization works very well for small numbers.
However, determining the prime factors of a very large integer can become computationally difficult.

Cryptographic Significance

The difficulty of factoring very large integers is an important idea in public-key cryptography.

Why Prime Numbers Matter in Cryptography

Prime numbers are important because they provide useful mathematical properties for cryptographic algorithms.
They appear in concepts such as:

- Modular arithmetic
- Public-key cryptography
- Key generation
- Multiplicative inverses
- Euler's theorem
- RSA-type constructions

Prime Numbers as Building Blocks

Every positive integer can be viewed as a combination of prime numbers.
For example:

$$3600 = 2^4 \times 3^2 \times 5^2$$

Hence, primes can be thought of as:

“Atoms of the Integers”

because composite integers are built by multiplying primes together.

Think and Answer

If:

$$a = 2^3 \times 3^2 \times 5$$

and:

$$b = 2^4 \times 3 \times 5^2$$

answer the following:

1. What is ab?
2. Does $a \mid b$?
3. What is $\gcd(a, b)$?

Solution

Given:

$$a = 2^3 \times 3^2 \times 5$$

$$b = 2^4 \times 3 \times 5^2$$

Product:

$$ab = 2^7 \times 3^3 \times 5^3$$

For divisibility:

$$3^2 \nmid 3^1$$

therefore:

$$a \nmid b$$

GCD:

Quick Questions

Answer briefly:

1. What is a prime number?
2. Is 1 a prime number?
3. What is the only even prime?
4. State the fundamental theorem of arithmetic.
5. What is the prime factorization of 60?
6. What condition must hold for $a \mid b$?
7. How are prime exponents combined during multiplication?
8. How are prime exponents used to compute GCD?

Descriptive Questions

1. Define a prime number and explain its importance in cryptography.
2. State and explain the fundamental theorem of arithmetic.
3. Explain the representation of an integer using prime exponents.
4. Show how multiplication can be performed using prime-power representations.
5. Explain divisibility using prime-factor exponents.
6. Explain how GCD can be computed using prime factorization.

Lecture Summary

Today we studied:

- Divisibility
- Prime and composite numbers
- Fundamental theorem of arithmetic
- Unique prime factorization
- Prime-exponent representation
- Multiplication using exponents
- Divisibility using prime exponents
- Greatest common divisor
- Prime factorization and computational difficulty
- Importance of prime numbers in cryptography

Formula Recap

$$\boxed{a = \prod_{p \in P} p^{a_p}}$$

Multiplication:

$$\boxed{(ab)_p = a_p + b_p}$$

Divisibility:

$$\boxed{a \mid b \iff a_p \leq b_p \text{ for every } p}$$

GCD:

$$\boxed{\gcd(a, b)_p = \min(a_p, b_p)}$$

References

- William Stallings, *Cryptography and Network Security: Principles and Practice*.
- Number Theory and Public-Key Cryptography section:
  - Prime Numbers
  - Prime Factorization
  - Divisibility
  - Greatest Common Divisor

Questions & Discussion

Prime numbers are simple to define,

but they form the mathematical foundation
of many powerful cryptographic systems.

Your Feedback Matters!

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